

Markscheme

Mathematics: analysis and approaches

Practice question bank

22. (a)

Consider the function $f(x) = \frac{2x + 61}{(x - 7)(x + 8)}$.

Write $f(x)$ as the sum of partial fractions in the form $\frac{A}{x - 7} + \frac{B}{x + 8}$.

attempt to write $2x + 61 = A(x + 8) + B(x - 7)$ (M1)

$2x + 61 = A(x + 8) + B(x - 7)$ A1

substituting $x = 7$: $75 = 15A \Rightarrow A = 5$ A1

substituting $x = -8$: $45 = -15B \Rightarrow B = -3$ A1

[4 marks]